

Autodesk® Scaleform®

Scaleform 4.2

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:
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Autodesk® Scaleform® 4.2

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목차

1	서문	4
2	마우스 클래스 확장	3
2.1		3
2.2		3
2.3	Mouse	5
2.4		7
2.5		9
3	Button 클래스 확장.....	13
4	MovieClip 클래스 확장	15
5	Netstream 클래스 확장	19
6	Selection 클래스 확장.....	23
7	TextField 클래스 확장.....	33
7.1		33
7.2		43
7.3		50
7.4	HTML	53
7.5	Shadow	55
7.6		59
7.7	IME	63
8	TextFormat 클래스 확장.....	66
9	Stage 클래스 확장.....	67
10	배열 클래스 확장	70
11	문자열 클래스 확장.....	71
12	Video 클래스 확장.....	72
13	전역 확장.....	73

14 3Di 클래스 확장.....76

15 표준 메소드 및 이벤트 핸들러 익스텐션77

1 서문

Autodesk Scaleform SDK Flash®

PC . Scaleform

, , .

, IME .

Scaleform " "

. Scaleform Player _global.gfxExtensions

true .

_global.gfxExtensions = true;

FLA .

Scaleform Player

.

Scaleform . Scaleform

SDK .

Scaleform Player .

Scaleform API

Scaleform API ,

.

2 마우스 클래스 확장

```

        Scaleform
        RIGHT, MIDDLE ( 16 )
        . Scaleform
        . Flash Mouse Class
        .

```

2.1

```

Wii . Scaleform
3.2 4 . Gfx::Movie::SetMouseCursorCount(unsigned n)
        . Gfx::MouseEvent Gfx::Movie::HandleEvent
Scaleform . Gfx::MouseEvent
Gfx::MouseEvent 0 .

```

2.2

```

onRollOver, onRollOut, onDragOver, onDragOut, onPress, onRelease, onReleaseOutside
        Scaleform .
        0 . onPress
        1 1 .

mc.onPress = function(mouseIdx:Number)
{
    if (mouseIndex == 0)
        . . .
    else if (mouseIndex == 1)
        . . .
    . . .
}

```

```

onPress    onRelease
.          1,    /    0
.
.

mc.onPress = function(mouseIdx:Number, keyboardOrMouse:Number)
{
    if (keyboardOrMouse == 0)
        . . .
    else
        . . .
    . . .
}

    onRollOver/Out    onDragOver/Out
rollover/dragover
.
    onRollOver/onRollOut    onDragOver/onDragOut
. onRollOver/onDragOver
roll
over    onRollOut/onDragOut
.
,    onRollOver/onRollOut
onDragOver/onDragOut    (
0    .    onRollOver/onDragOver
0
onRollOver/onRollOut    1
    onRollOut/onDragOut    1
onRollOut/onDragOut    0
,    onRollOver/onRollOut    onDragOver/onDragOut
.

mc.onRollOver = function(mouseIdx:Number, nestingIdx:Number)
{
    // do roll over animation only if nestingIdx == 0
    if (nestingIdx == 0)
        doOverAnimation();
    . . .
}
mc.onRollOut = function(mouseIdx:Number, nestingIdx:Number)
{

```



```

        // do roll out animation only if nestingIdx == 0
        if (nestingIdx == 0)
            doOutAnimation();
        . . .
    }

```

On(rollOver), on(rollOut)

onPress, onRelease

- onPressAux
- onReleaseAux
- onReleaseOutsideAux
- onDragOver
- onDragOut

, index != 0

(index == 0).

```

mc.onPressAux = function(mouseIdx:Number, keyboardOrMouse:Number, buttonIdx:Number)
{
    if (buttonIdx == 1) //
        . . .
    . . .
}

```

2.3 Mouse

- `onMouseMove = function() { }`
- `onMouseDown = function() { }`
- `onMouseUp = function() { }`
- `onMouseWheel = function(delta : Number, targetPath : String) { }`

```
(
    _global.gfxExtensions    true
    Scaleform
```

Scaleform RIGHT, MIDDLE

LEFT

6

```

    Mouse.addListener(mouseListener);

    /
    (
    .
    Mouse["LEFT"], Mouse["RIGHT"]
    Mouse["MIDDLE"] (Mouse["LEFT"]
    2.0 Mouse.LEFT ).
    .
    'undefined'
    'undefined'
    . 'undefined' LEFT
    .
    .

    ▪ onMouseMove = function([mouseIdx : Number],
        [x : Number], [y : Number]) { }
    ▪ onMouseDown = function(button : Number, [targetPath : String],
        [mouseIdx : Number], [x : Number], [y : Number],
        [dblClick : Boolean]) { }
    ▪ onMouseUp = function(button : Number, [targetPath : String],
        [mouseIdx : Number], [x : Number], [y : Number]) { }
    ▪ onMouseWheel = function(delta : Number, targetPath : String,
        [mouseIdx : Number], [x : Number], [y : Number]) { }

    mouseIdx 0
    x y .(_root ).
    onMouseDown dblClick . true

```

2.4

- *Arrow* -
- *Hand* - useHandCursor true
- *I-Beam* -

, Scaleform

C++ API

. C++ Gfx::StateBag::SetUserEventHandler

FxPlayer.cpp

SDK

Mouse fla/swf

'Cursor_M'

"hand", "arrow" 'ibeam"

_x _y

Scaleform

Mouse.setCursorType

cursorType mouseX

cursorType

(Mouse["ARROW"], Mouse["HAND"],

Mouse["IBEAM"])

```
_global.gfxExtensions = 1;
// Hide system cursor.
Mouse.hide();

var mouseListener:Object = new Object;
mouseListener.onMouseMove = function(mouseIdx, x,y)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    if (x == undefined)
    {
        x = _xmouse;
        y = _ymouse;
    }
}
```

```

    }
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));

    cmc._x = x;
    cmc._y = y;
}
Mouse.addListener(mouseListener);

Mouse["setCursorType"] = function(cursorType, mouseIdx)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));
    switch(cursorType)
    {
        case Mouse["HAND"]:
            cmc.Cursor.gotoAndPlay("hand");
            break;
        case Mouse["ARROW"]:
            cmc.Cursor.gotoAndPlay("arrow");
            break;
        case Mouse["IBEAM"]:
            cmc.Cursor.gotoAndPlay("ibeam");
            break;
        default: return;
    }
}

// Override Mouse.show and Mouse.hide functions so
// that they affect our cursor.
ASSetPropFlags(Mouse, "show,hide", 0, 7);

Mouse.show = function(mouseIdx)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));
    cmc._visible = true;
}
Mouse.hide = function(mouseIdx)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));
    cmc._visible = false;
}

```

2.5

Mouse

GFx

. 1.0 Mouse.setCursorType(...)

2.0

2.0

: Mouse["setCursorType"](...).

getButtonsState() 정적 메소드

```
public function getButtonsState(mouseIndex : Number) : Number
```

Scaleform version: 2.2

0

Parameters

mouseIndex : Number – 0

getTopMostEntity() 정적 메소드

```
public function getTopMostEntity([mouseIndex : Number]) : Object  
public function getTopMostEntity(testAll : Boolean,  
                                  [mouseIndex : Number]) : Object  
public function getTopMostEntity(x : Number, y : Number) : Object  
public function getTopMostEntity(x : Number, y : Number, testAll: Boolean) : Object
```

Scaleform version: 1.2.34, 2.2

(onPress, onMouseDown, onRollOver)

```
MovieClip.hitTest . hitTest x,y  
    , getTopMostEntity x,y
```

```
Mouse.getTopMostEntity(x, y).hitTest(x, y, true) == true.
```

Parameters

mouseIndex : Number – 0

```
testAll : Boolean - (false) (true) .
getTopMostEntity 'true' .
```

```
x : Number, y : Number -
```

사용예:

```
var listenerObj = new Object;
listenerObj.onMouseMove = function()
{
    var target = Mouse["getTopMostEntity"]();
    trace("Mouse moved, target = "+target);
}
Mouse.addListener(listenerObj);
var target = Mouse["getTopMostEntity"](480, 10);
trace("The character at the (480, 10) is " + target);
```

getPosition 정적 메소드

```
public function getPosition(mouseIndex : Number) : flash.geom.Point
```

Scaleform version: 2.2

```
_root .
flash.geom.Point .
```

Parameters

```
mouseIndex : Number - 0
```

setCursorType() 정적 메소드

```
public function setCursorType(cursorType : Number, [mouseIndex : Number]) : void
```

Scaleform version: 1.2.32

```
cursorType . "
```

Parameters

`cursorType` : Number – Mouse.ARROW, Mouse.HAND, Mouse.IBEAM

`mouseIndex` : Number – 0

3 Button 클래스 확장

hitTestDisable 속성

hitTestDisable:Boolean [read-write]

Scaleform version: 2.1.51

```
true      MovieClip.hitTest
false     .
:
TextField.hitTestDisable
MovieClip.hitTestDisable
```

topmostLevel 속성

topmostLevel:Boolean [read-write]

Scaleform version: 2.1.50

```
true
false
"topmostLevel "
:
• Scaleform 3.0.71 "topmostLevel "
true . ( "topmostLevel"
.)
• Scaleform 3.0.72
: ObjectA ObjectB "topmostLevel"
```


4 MovieClip 클래스 확장

hitTest () 메소드

```
public hitTest(x:Number, y:Number, [shapeFlag:Boolean],
               [ignoreInvisibleChildren:Boolean]):Boolean
```

Scaleform version: 2.1.51

```
3          hitTest
          . hitTest
          ( , _visible false _alpha 0 invisible
            ) "true" .
```

Parameters

ignoreInvisibleChildren:Boolean – true

hitTestDisable 속성

hitTestDisable:Boolean [read-write]

Scaleform version: 1.2.32

```
true      MovieClip.hitTest
false     .
false     .
:
TextField.hitTestDisable
Button.hitTestDisable
```

noAdvance 속성

noAdvance : Boolean

Scaleform version: 2.1.52

```
true                                Advance    off    .
                                   .
                                   advance    (
                                   ).          "true"    Scaleform
                                   .
                                   :
                                   _global.noInvisibleAdvance
```

topmostLevel 속성

topmostLevel:Boolean [read-write]

Scaleform version: 2.1.50

```
    true                                .
                                   .
    false    .
                                   "topmostLevel "    ,    :
• Scaleform 3.0.71    "topmostLevel "
    true                                . (    "topmostLevel"
                                   . )
• Scaleform 3.0.72
    .    :    ObjectA    ObjectB    "topmostLevel"
true    ObjectA    ObjectB
    .
    :    "topmostLevel"    swapDepth
    .
    :
```

```
TextField.topmostLevel  
Button.topmostLevel
```

rendererString 속성

rendererString:String [read-write]

Scaleform version: 2.2.55

MovieClip

예:

```
myMovieInstance.rendererString = "SHADER_Blur"
```

rendererFloat 속성

rendererFloat:Number [read-write]

Scaleform version: 2.2.55

MovieClip

Float

예:

```
myMovieInstance.rendererFloat = 4; // eg: C++ enum value
```

disableBatching 속성

disableBatching:Boolean

Scaleform version: 4.0.17

focusGroupMask 프로퍼티

focusGroupMask : Number

Scaleform version: 3.3.84

모든
focus group , controller focus
. Focus group setController FocusGroup
controller .
 , "button1" controller 0 1 focus
focus group controller .

```
Selection.setControllerFocusGroup(0, 0);  
Selection.setControllerFocusGroup(1, 1);
```

```
button1.focusGroupMask = 0x1; // bit 0 - focus group 0  
movieclip2.focusGroupMask = 0x1 | 0x2 // bits 0 and 1 - focus groups 0 and 1
```

"focusGroupMask" movieclip .

5 Netstream 클래스 확장

Scaleform NetStream

onMetaData 이벤트 핸들러

```
onMetaData = function(info:Object) { }
```

Scaleform version: 3.0.63

, 2

onMetaData 2

1. audioTracks –

audioTracks

channelsNumber – (1-mono, 2- stereo, 6 – 5.1 surround sound)

totalSamples –

trackIndex – NetStream.audioTrack
.()

sampleRate –

2. subtitleTracksNumber –

audioTrack 속성

audioTrack:Number [read-write]

Scaleform version: 3.0.63

set/get

사용예:

```
var ns:NetStream = new NetStream(nc);
var audioTracks;

ns.onMetaData = function(info:Object)
{
    audioTracks = info.audioTracks;
}

if (audioTracks != undefined && audioTracks.length > 0)
    ns.audioTrack = audioTracks[0].trackIndex;
```

subtitleTrack 속성

subtitleTrack:Number [read-write]

Scaleform version: 3.0.63

set . 0 .

onSubtitle 이벤트 핸들러

```
onSubtitle = function(msg:String) {}
```

Scaleform version: 3.0.63

예:

```
var ns:NetStream = new NetStream(nc);
var subtitlesNumber = 0;

ns.onMetaData = function(info:Object)
{
    subtitlesNumber = info.subtitleTracksNumber;
}

ns.onSubtitle = function(msg:String)
{
    sbTextField.text = msg;
}

if (subtitlesNumber > 0)
    ns.subtitleTrack = 1;
```


setNumberOfFramePools 함수

```
public function setNumberOfFramePools(pools : Number) : Void
```

Scaleform version: 3.0.68

```
        .
        , "frame pool"
        . CPU
        , frame
pool
        . 1
        .
        .( , NetStream.play()
        ).
```

setReloadThresholdTime 함수

```
public function setReloadThresholdTime(reloadTime : Number) : Void
```

Scaleform version: 3.0.68

```
        . Scaleform Video
        threshold(
threshold(
        ) bitrate
        , reload
        . reload
        0.8(
        .
        , Netstream
        2
        (NetStream.setBufferTime()
        ) threshold(
        1
        ,
1
        .
        .
        ,
        .
        .( , NetStream.play()
        )
```


6 Selection 클래스 확장

Selection

Scaleform Selection . 1.0

Selection.captureFocus() Selection .

2.0

. ActionScript 2.0 .

Selection["captureFocus"]().

alwaysEnableArrowKeys 정적 속성

alwaysEnableArrowKeys : Boolean

Scaleform version: 1.2.34

```
        "_focusrect"        false
    (
    ).        "_focusrect = false;"
        alwaysEnableArrowKeys    "true" .
```

예:

```
Selection["alwaysEnableArrowKeys"] = true;
```

alwaysEnableKeyPress 정적 속성

alwaysEnableKeyPress : Boolean

Scaleform version: 3.0.63

"_focusrect" false(,)

onPress/ onRelease . "_focusrect = false;"

`alwaysEnableKeyPress` `true` .

예:

```
Selection["alwaysEnableKeyPress"] = true;
```

captureFocus()정적 메소드

`public function captureFocus([doCapture:Boolean, controllerIdx:Number]) : void`

Scaleform version: 1.2.34

`doCapture` `true` () `captureFocus`

`doCapture` `false` `captureFocus`
on

Parameters

`doCapture:Boolean` - , / .

`captureFocus` `true` .

`controllerIdx:Number` - , capture / .
controller 0 .

예:

```
Selection["captureFocus"](); // same as passing "true"  
Selection["captureFocus"](false);
```

disableFocusAutoRelease 정적속성

`disableFocusAutoRelease` : Boolean

Scaleform version: 1.2.34

```

(
    ).
)
true
:
Selection["disableFocusAutoRelease"] = true;

```

disableFocusKeys 정적 속성

disableFocusKeys : Boolean

Scaleform version: 2.2.58

(TAB, Shift-TAB).

예:

```
Selection["disableFocusKeys"] = true;
```

```
moveFocus()
```

disableFocusRolloverEvent 정적 속성

disableFocusRolloverEvent : Boolean

Scaleform version: 2.0.37

/

/

true

:

```
Selection["disableFocusRolloverEvent "] = true;
```

modalClip 정적 속성

modalClip : MovieClip

Scaleform version: 2.2.58

```
                "modal"
(TAB, Shift-TAB )
                "
                "
                .
```

예:

```
Selection["modalClip"] = _root.mc;
...
Selection["modalClip"] = undefined;
```

```
disableFocusKeys
moveFocus()
```

moveFocus() static 메소드

```
public function moveFocus(keyToSimulate : String [, startFromMovie:Object,
    includeFocusEnabledChars : Boolean = false,
    controllerIdx :Number]) : Object
```

Scaleform version: 2.2.58

```
(TAB, Shift-TAB )
                .disableFocusKeys modalClip
                .
```

Parameters

keyToSimulate:String – "up", "down", "left", "right", "tab", "shifftab"

startFromMovie:Object –

moveFocus . null

undefined .

includeFocusEnabledChars:Boolean –focusEnabled ,

tabEnabled/tabIndex moveFocus

0

tabEnabled/tabIndex

controllerIdx:Number – , capture /
controller 0 .

예:

```
Selection["moveFocus"]("up");  
Selection["moveFocus"]("tab", _root.mc);  
Selection["moveFocus"]("tab", _root.mc, true);  
Selection["moveFocus"]("tab", null, true);
```

반환값

undefined .

disableFocusKeys
modalClip

findFocus() 정적 메소드

```
public function findFocus(keyToSimulate : String [, parentMovie:Object, loop :
    Boolean, startFromMovie:Object, includeFocusEnabledChars : Boolean,
    controllerIndex : Number]) : Object
```

Scaleform version: 3.3.84

```
TAB, Shift + TAB
focus
.disableFocusKeys
.setModalClip/getModalClip
.focus
```

파라미터

```
keyToSimulate:String - ("up", "down", "left", "right", "tab",
    "shift + tab")
parentMovie - modal clip movie clip . focus
clip . null
loop - focus Boolean
    . , focus
    "down" , findFocus "null"( "false"
    ) focus (
    "true" )
startFromMovie:Object - focus findFocus
    null focus
includeFocusEnabledChars:Boolean - tabEnabled/tabIndex
    focusEnabled
    moveFocus false
    tabEnabled/tabIndex
    focus
controllerIndex - focus controller zero base index
    focus group controller focus
```

예:


```
var a = Selection["findFocus"]("up");
```

반환

, focus null

```
:
  disableFocusKeys
  setModalClip
  getModalClip
```

setModalClip() 정적 메소드

```
public function setModalClip(modalClip : Object, controllerIndex : Number)
```

Scaleform version: 3.3.84

movie clip focus "modal" clip .
 TAB, Shift + TAB "TAB " movie clip
 focus .

파라미터

modalClip – A modal clip.
 controllerIndex - controller zero base index .

getModalClip() 정적 메소드

```
public function getModalClip(controllerIndex : Number) : Object
```

Scaleform version: 3.3.84

controller modal clip .

파라미터

controllerIndex - controller Zero base index .

반환

modal clip undefined .

setControllerFocusGroup() 정적 메소드

```
public function setControllerFocusGroup (controllerIndex : Number, focusGroupIdx :  
Number) : Boolean
```

Scaleform version: 3.3.84

	controllerIndex	controller	focus group	
	controller	focus group 0	,	focus
		controller	focus	.
	, controller	focus	(	)
setControllerFocusGroup (1,1)	controller 1	focus group	.	
setControllerFocusGroup (1,0)	controller 0	1	focus	

.

파라미터

controllerIndex	- controller	zero base index	.
focusGroupIdx	- focus group	zero base index	.

예:

```
Selection["setControllerFocusGroup"](0,0);  
Selection["setControllerFocusGroup"](1,1);  
Selection["setControllerFocusGroup"](2,1);
```

반환

true .

getControllerFocusGroup() 정적 메소드

```
public function getControllerFocusGroup (controllerIndex : Number) : Number
```

Scaleform version: 3.3.84

controller focus group index .

파라미터

controllerIndex - controller zero base index .

반환

focus group zero base index .

getFocusArray() 정적 메소드

```
public function getFocusArray(mc : Object) : Array
```

Scaleform version: 3.3.84

movie clip/button/text field focus controller
index .

파라미터

mc - movie clip, button text field.

반환

zero base index ().

getFocusBitmask() 정적 메소드

```
public function getFocusBitmask(mc : Object) : Number
```

Scaleform version: 3.3.84

movieclip/button/textfield , bit
focus controller .

파라미터

mc - movie clip, button text field.

반환

controller .

getControllerMaskByFocusGroup() 정적 메소드

```
public function getControllerMaskByFocusGroup (focusGroupIdx : Number) : Number
```

Scaleform version: 3.3.84

controller , focus group
. setControllerFocusGroup
.

파라미터

focusGroupIdx - focus group index .

반환

controller .

numFocusGroups 프로퍼티

numFocusGroups : Number

Scaleform version: 3.3.84

setControllerFocusGroup focus group . focus
group 0 3 , numFocusGroups 2 .

7 TextField 클래스 확장

```
Scaleform TextField
8 TextField
9
IME,
,
( , ),
,
.
```

7.1

autoFit 속성

autoFit:Boolean [read-write]

Scaleform version: 2.0.39

false .

appendText () 메소드

public function appendText(newText:String):void

Scaleform version: 2.0.37

```
newText
(+=)
( , my_txt.text += appendingText)
: style sheet text field
```

Parameters

newText:String -

:

appendHtml()

appendHtml () 메소드

```
public function appendHtml(newHtml:String):void
```

Scaleform version: 2.0.37

```
newHtml          HTML
htmlText          (+=)          ( , txt.htmlText += moreHtml).
htmlText          += HTML          HTML
HTML              .          HTML , newHtml
HTML              ( newHtml HTML +=
                  ).
: style sheet text field .
```

Parameters

newHtml:String - HTML

:

appendText()

caretIndex 속성

caretIndex:Number [read-only]

Scaleform version: 2.0.37

```
        (
    )
    (
    )caret
    caret
    .
    0
    Selection.getCaretIndex()
    . caretIndex
    Selection.getCaretIndex() -1

    Caret
    0
    (
    0
    ).
```

```
Selection.getCaretIndex()
```

getCharBoundaries () 메소드

```
public function getCharBoundaries(charIndex:Number): flash.geom.Rectangle
```

Scaleform version: 2.0.37

```

    .
    ,
    .
    (
    ,
    'a', 'w' 'k'
    ).
```



Parameters

```
    charIndex:Number -
    0
    (
    0,
    1)
```

반환값

flash.geom.Rectangle - x y ,
 , (0,0) .

getExactCharBoundaries()

getExactCharBoundaries () 메소드

public function getExactCharBoundaries(charIndex:Number): flash.geom.Rectangle

Scaleform version: 2.0.37

(getCharBoundaries()).

'a', 'w' 'k'



Parameters

charIndex:Number - 0 (0, 1)

반환값

flash.geom.Rectangle - x y ,
 , (0,0) .

getCharBoundaries()

getCharIndexAtPoint () 메소드

```
public function getCharIndexAtPoint(x:Number, y:Number): Number
```

Scaleform version: 2.0.37

x y (0).

Parameters

x:Number – x

y:Number – y

반환값

Number - 0 (0, 1). -1 .

getFirstCharInParagraph () 메소드

```
public function getFirstCharInParagraph(charIndex:Number): Number
```

Scaleform version: 2.0.37

charIndex .

Parameters

charIndex:Number - 0 (0, 1)

반환값

Number – 0

getLineIndexAtPoint () 메소드

```
public function getLineIndexAtPoint(x:Number, y:Number): Number
```

Scaleform version: 2.0.37

x y 0
.

Parameters

x:Number – x
y:Number – y

반환값

Number - 0 (0,
1). -1 .

getLineLength() 메소드

```
public function getLineLength(lineIndex:Number): Number
```

Scaleform version: 2.0.37

.

Parameters

lineIndex:Number - 0 (0,
1)

반환값

Number –

getLineMetrics() 메소드

```
public function getLineMetrics(lineIndex:Number): Object
```

Scaleform version: 2.0.43

ascent : Number

ascent

descent : Number

descent

height : Number

()

leading : Number

width : Number

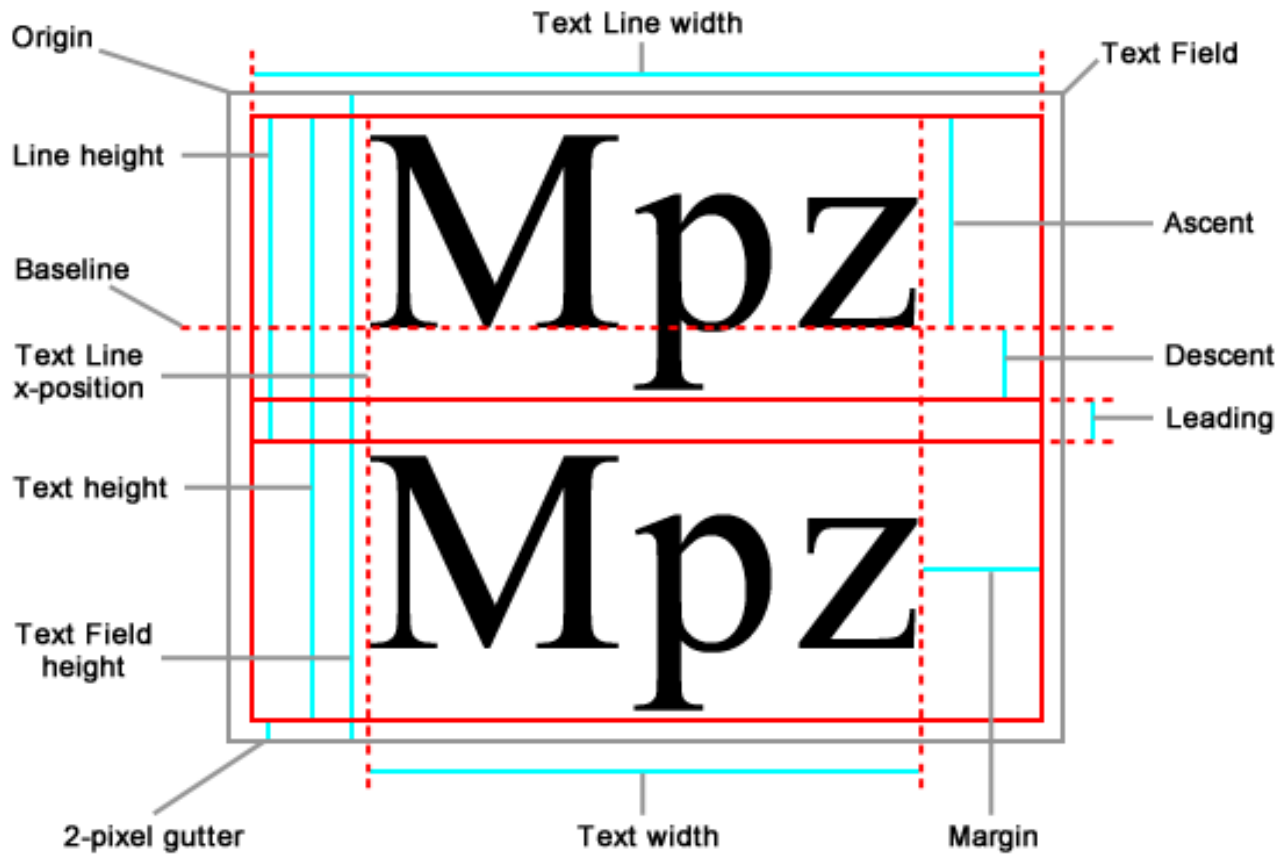
width

(

).

x : Number

x



Parameters

lineIndex: Number - 0 (0, 1)

반환값

Number -

getLineOffset () 메소드

```
public function getLineOffset(lineIndex: Number): Number
```

Scaleform version: 2.0.37

lineIndex

Parameters

```
lineIndex:Number - 0 (0, 1)
```

반환값

Number - 0

getText () 메소드

```
public function getText(lineIndex:Number): String
```

Scaleform version: 2.0.43

lineIndex .

Parameters

```
lineIndex:Number - 0 (0, 1)
```

반환값

String -

hitTestDisable 속성

```
hitTestDisable:Boolean [read-write]
```

Scaleform version: 1.2.32

true MovieClip.hitTest . ,

false .

noTranslate 속성

```
noTranslate:Boolean [read-write]
```

```
true      GfXTranslator::Translate      .
false    .
```

numLines: Number [read-only]

```
wrap . wordWrap true
```

topmostLevel: Boolean [read-write]

```

true
.

false
.
```

```

• Scaleform 3.0.71      "topmostLevel "
      true                . (      "topmostLevel"
                           . )

```

```

true      .      :      ObjectA      ObjectB      "topmostLevel"
           .      :      ObjectA      ObjectB
           .      :

```

: "topmostLevel" swapDepth

:
MovieClip.topmostLevel
Button.topmostLevel

focusGroupMask 프로퍼티

focusGroupMask : Number

Scaleform version: 3.3.84

모든
focus group , controller focus
. Focus group setController FocusGroup
controller
 , "button1" controller 0 1 focus
focus group controller

```
Selection.setControllerFocusGroup(0, 0);  
Selection.setControllerFocusGroup(1, 1);
```

```
button1.focusGroupMask = 0x1; // bit 0 - focus group 0  
movieclip2.focusGroupMask = 0x1 | 0x2 // bits 0 and 1 - focus groups 0 and 1
```

"focusGroupMask" movieclip

7.2

alwaysShowSelection 속성

alwaysShowSelection:Boolean [read-write]

Scaleform version: 2.1.45

```

        Scaleform
        true
        Scaleform
        (" "
        ).
        false
        :
        selectionBkgColor
        selectionTextColor
        inactiveSelectionBkgColor
        inactiveSelectionTextColor
```

copyToClipboard () 메소드

```
public function copyToClipboard([richTextClipboard:Boolean], [startIndex:Number],
[endTime:Number]):void
```

Scaleform version: 2.1.45

Parameters

```

    richClipboard:Boolean – true
    useRichTextClipboard
    startIndex:Number –
    selectionBeginIndex
    endIndex:Number –
    selectionEndIndex
    :
```



```

useRichTextClipboard
selectionBeginIndex
selectionEndIndex
cutToClipboard()
pasteFromClipboard()

```

cutToClipboard () 메소드

```

public function cutToClipboard([richClipboard:Boolean], [startIndex:Number],
[endIndex:Number]):void

```

Scaleform version: 2.1.45

Parameters

richClipboard:Boolean - true

useRichTextClipboard

startIndex:Number -

selectionBeginIndex

endIndex:Number -

selectionEndIndex

:

```

useRichTextClipboard
selectionBeginIndex
selectionEndIndex
copyToClipboard()
pasteFromClipboard()

```

inactiveSelectionBkgColor 속성

inactiveSelectionBkgColor:Number [read-write]

Scaleform version: 2.1.45

```
        . alwaysShowSelection      true
        . 0xAARRGGBB              , AA      [0...255]    16      , RR
        [0...255]    16      , GG      [0...255]    16      , BB
[0...255]    16      .
        0
        .
        . 0xFFFF0000(
0xFF(255))      backgroundColor      0xFF0000      .
:
alwaysShowSelection
inactiveSelectionTextColor
selectionBkgColor
selectionTextColor
```

inactiveSelectionTextColor 속성

inactiveSelectionTextColor: Number [read-write]

Scaleform version: 2.1.45

```
        . alwaysShowSelection      true
        . 0xAARRGGBB              , AA      [0...255]    16      , RR
        [0...255]    16      , GG      [0...255]    16      , BB
[0...255]    16      .
        0
        .
        . 0xFFFF0000(
0xFF(255))      textColor      0xFF0000      .
:
alwaysShowSelection
```

inactiveSelectionBkgColor
selectionBkgColor
selectionTextColor

noAutoSelection 속성

noAutoSelection:Boolean [read-write]

Scaleform

selectionBeginIndex 속성

selectionBeginIndex:Number [read-only]

Scaleform version: 2.1.45

```
0
caretIndex
Selection.getBeginIndex() selectionBeginIndex
Selection.getBeginIndex() -1
:
caretIndex
selectionEndIndex
Selection.getBeginIndex()
```

selectionEndIndex 속성

selectionEndIndex:Number [read-only]

Scaleform version: 2.1.45

```
0
caretIndex
Selection.getEndIndex() selectionEndIndex
Selection.getEndIndex() -1
:
caretIndex
selectionBeginIndex
Selection.getEndIndex()
```

selectionBkgColor 속성

selectionBkgColor: Number [read-write]

Scaleform version: 2.1.45

```
        . alwaysShowSelection      true
        . 0xAARRGGBB               , AA      [0...255]    16      , RR
        [0...255]    16      , GG      [0...255]    16      , BB
[0...255]    16      .
        0
        .
        0xFFFF0000(
0xFF(255))      backgroundColor      0xFF0000
:
inactiveSelectionTextColor
inactiveSelectionBkgColor
selectionTextColor
```

selectionTextColor 속성

selectionTextColor: Number [read-write]

Scaleform version: 2.1.45

```
        . alwaysShowSelection      true
        . 0xAARRGGBB               , AA      [0...255]    16      , RR
        [0...255]    16      , GG      [0...255]    16      , BB
[0...255]    16      .
        0
        .
        0xFFFF0000(
0xFF(255))      textColor      0xFF0000
.
```

```

        :
        inactiveSelectionBkgColor
        inactiveSelectionBkgColor
        selectionBkgColor

```

useRichTextClipboard 속성

useRichTextClipboard:Boolean [read-write]

Scaleform version: 2.1.45

```

        . True
        Scaleform
        ( , ).
        useRichTextClipboard true
        false

```

7.3

```

        .
        ( ,
        ) Scaleform
        .

```

fontScaleFactor 속성

fontScaleFactor:Number [read-write]

Scaleform version: 2.0.43

```

        .
        fontScaleFactor
        1.0

```

textAutoSize 속성

textAutoSize:String [read-write]

Scaleform version: 2.0.43

textAutoSize

"none", "shrink" "fit" ("shrink" "fit")

(5pt)

textAutoSize "fit"

"shrink"



"none"

verticalAlign 속성

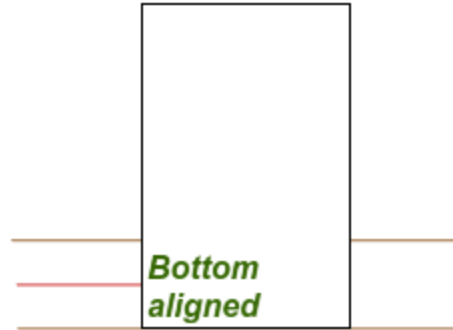
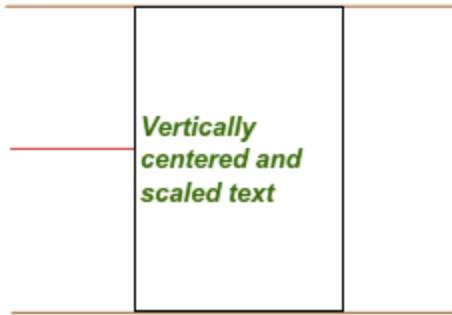
verticalAlign:String [read-write]

Scaleform version: 2.0.43

"none" ("none"

"top"), "bottom", "center" "center"

, "bottom" ().



"none" .

:

verticalAutoSize
TextField.align

verticalAutoSize 속성

verticalAutoSize:String [read-write]

Scaleform version: 2.0.43

verticalAutoSize "none" (), "top",
"bottom", "center" . verticalAutoSize "none" ()

verticalAutoSize "top" wordWrap true
TextFieldAutoSize ().

verticalAutoSize "center" .
().

verticalAutoSize "bottom" .
().



"none" .

:
verticalAlign
TextField.autoSize

7.4 HTML

Scaleform HTML

 태그, ALPHA 속성

ALPHA="#xx"

00 FF (0 - 255). 0 , 255
. COLOR ALPHA HTML

 태그

Scaleform 2.0/2.1

. Scaleform

("http://", "file://")

. 5.6

.

.

7.5 Shadow

(,) .

blurX, blurY 속성

blurX:Number [read-write]

blurY:Number [read-write]

Scaleform version: 2.0.41

. 0 15.9 ().

0 .

:

blurStrength

blurStrength 속성

blurStrength:Number [read-write]

Scaleform version: 2.0.41

. 0 15.9 ().

0 .

:

blurX

blurY

shadowAlpha 속성

shadowAlpha:Number [read-write]

Scaleform version: 2.0.41

. 0.00 1.00 . .25
25% .

:
shadowColor

shadowAngle 속성

shadowAngle:Number [read-write]

Scaleform version: 2.0.41

DropShadowFilter . 0 360° (
) . 45 .

:
DropShadowFilter

shadowBlurX, shadowBlurY 속성

shadowBlurX:Number [read-write]
shadowBlurY:Number [read-write]

Scaleform version: 2.0.41

. 0 15.9 ().
0 .

:
shadowStrength

shadowColor 속성

shadowColor:Number [read-write]

Scaleform version: 1.1.31

(shadowStyle). 0xRRGGBB .
RR 16 [0..255], BB 16 [0..255] GG
16 [0..255] .

shadowStyle

shadowDistance 속성

shadowDistance:Number [read-write]

Scaleform version: 2.0.41

DropShadowFilter .

:
DropShadowFilter

shadowHideObject 속성

shadowHideObject:Boolean [read-write]

Scaleform version: 2.0.41

. True .
false () .

shadowKnockOut 속성

shadowKnockOut:Boolean [read-write]

Scaleform version: 2.0.41

```
(true)      .
.      false (      )      .
```

shadowQuality 속성

shadowQuality: Number [read-write]

Scaleform version: 2.0.41

1 2 .1

shadowStrength 속성

shadowStrength: Number [read-write]

Scaleform version: 2.0.41

```

0      .      0      15.9      (      ).

0      .

:

shadowBlurX
shadowBlurY

```

shadowStyle 속성

```
shadowStyle:String [read-write]
```

Scaleform version: 1.1.31

```
shadowColor . shadowStyle
```

```
field.shadowStyle = "s{0,-1.5}{-1.4,1.2}{1.4,1.2}t{0,0}";
field.shadowColor = 0xff0000;
```

```
    "s" shadowColor
    "t" TextField.textColor
    (delimiter)
    . "t"
    "t{0,0}"
```

draw primitive

```
    :
    shadowColor
    TextField.textColor
```

7.6

setImageSubstitutions () 메소드

```

public setImageSubstitutions(substInfoArr:Array) : void
public setImageSubstitutions(substInfo:Object) : void
public setImageSubstitutions(null) : void

```

Scaleform version: 2.0.38

SWF

SWF

- 1.
2. () ()CTRL
- Linkage
3. Export for ActionScript , (, myImage)
4. OK

BitmapData

```

import flash.display.BitmapData;
var imageBmp:BitmapData = BitmapData.loadBitmap("myImage");

```

주: !(import flash.display.BitmapData;
flash.display.BitmapData).
"undefined"

ID

subString:String

image : BitmapData

width : Number

height : Number

baseLineY : Number

Y

(

)

id : String

"updateImageSubstitution"

id

"substInfoArr"

"substInfo"

setImageSubstitutions(substInfo:Object)

setImageSubstitutions(substInfoArr:Array)

setImageSubstitutions

setImageSubstitutions(null)

setImageSubstitutions

Parameters

substInfoArr:Array -

()

substInfo:Object -

().

null -

:

```
updateImageSubstitution()
```

```
:
```

```
var b1 = BitmapData.loadBitmap("smile1");
var b2 = BitmapData.loadBitmap("smile2");
var b3 = BitmapData.loadBitmap("smile3");
var a = new Array;
a[0] = { subString:"=", image:b1, baseLineY:35, width:20, height:20, id:"sm=" } };
a[1] = { subString":-)", image:b2, baseLineY:20, id:"sm:-)" } };
a[2] = { subString":\\", image:b3, baseLineY:35, height:100 } };
a[3] = { subString":-\\", image:b1 } };
t.setImageSubstitutions(a);
```

"=)"

"smile1"

.

updateImageSubstitution () 메소드

```
public updateImageSubstitution(id:String, image:BitmapData) : void
```

Scaleform version: 2.0.38

setImageSubstitutions

.

Parameters

id:String - ID, setImageSubstitutions

id

image:BitmapData - . Null

:

setImageSubstitutions()

:

```
t.updateImageSubstitution("sm=", b3);
```

. onEnterFrame

setInterval

. updateImageSubstitution

. ,

.

```

var phase = 0;
var b1a = BitmapData.loadBitmap("smile1a");
var b2a = BitmapData.loadBitmap("smile2a");

onEnterFrame = function()
{
    if (phase % 10 == 0)
    {
        if (phase % 20 == 0)
        {
            _root.t.updateImageSubstitution("sm=", b1);
            _root.t.updateImageSubstitution("sm:-)", b2);
        }
        else
        {
            _root.t.updateImageSubstitution("sm=", b1a);
            _root.t.updateImageSubstitution("sm:-)", b2a);
        }
    }
    ++phase;
}

```

7.7 IME

IME

disableIME 속성

disableIME:Boolean [read-write]

Scaleform version: 2.1.50

True

IME

false

getIMECompositionStringStyle () 메소드

public function getIMECompositionStringStyle(categoryName:String): Object

Scaleform version: 2.1.50

```
IME
        . categoryName
        setIMECompositionStringStyle
    }
```

Parameters

categoryName:String – IME (setIMECompositionStringStyle).

반환값

Object – (\).

```

:
setIMECompositionStringStyle
disableIME
```

setIMECompositionStringStyle () 메소드

```
public function setIMECompositionStringStyle(categoryName:String,
                                             styleDescriptor:Object): Number
```

Scaleform version: 2.1.50

```
IME
        . categoryName
```

```

"compositionSegment" –
"clauseSegment"–
"convertedSegment" –
"phraseLengthAdj" –
"lowConfSegment" –
```

```
styleDescriptor
    }
```

textColor : Number

backgroundColor : Number

underlineColor : Number

underlineStyle : String

"single"

"thick"

"dotted"

"ditheredSingle"

"ditheredThick"

0xRRGGBB

RR

16

[0..255],

BB

16

[0..255]

GG

` 16

[0..255]

Parameters

categoryName:String – IME

()

styleDescriptor:Object –

()

:

getIMECompositionStringStyle

disableIME

9 Stage 클래스 확장

```
Stage                                     Scaleform
    .                                     .
    "visibleRect", "safeRect"          "originalRect"
    , "onResize"
    .
    1.0      Stage.visibleRect
    2.0
ActonScript 2.0
Stage["visibleRect"]..
```

visibleRect 속성

visibleRect:flash.geom.Rectangle [read-only]

Scaleform version: 2.2.56

```

    .
    ,
    /
    .
    .
    :
    safeRect
    originalRect
```

safeRect 속성

safeRect:flash.geom.Rectangle [read-only]

Scaleform version: 2.2.56

```

    .
    GFX::Movie::SetSafeRect
    "visibleRect"
    .
    :
    visibleRect
```

originalRect

originalRect 속성

originalRect:flash.geom.Rectangle [read-only]

Scaleform version: 2.2.56

```

    SWF
    "Document properties"
    (600px by 400px)
    {0, 0,
{600, 400} }
:
visibleRect
safeRect
```

onResize 이벤트 핸들러

onResize = function([visibleRect:flash.geom.Rectangle]) {}

Scaleform version: 2.2.56

```

onResize
    Stage.visibleRect
:
visibleRect
```

translateToScreen() 메소드

public function translateToScreen(pt:Object): Point

Scaleform version: 3.3.84

Stage point

파라미터

pt:Object – point x, y Object .
flash.geom.Point Object .

반환

Point – point .

10 배열 클래스 확장

로케일 지정 정렬

```
Array.sort      Array.sortOn
// 배열을 정렬하는 메서드. 정렬 기준을 지정할 수 있다.
// (umlaut, accent, acute) z
['á', 'z', 'a']      ['a', 'z', 'á']
// 배열을 정렬하는 메서드. 정렬 기준을 지정할 수 있다.
// (umlaut, accent, acute) z
['a', 'á', 'z']

Array.LOCALE:      Scaleform      Array.sort/sortOn
// 배열을 정렬하는 메서드. 정렬 기준을 지정할 수 있다.
// (umlaut, accent, acute) z
Array.LOCALE

var arr = ['á', 'z', 'a'];
var newArr1 = arr.sort(); // returns ['a', 'z', 'á']
var newArrLoc = arr.sort(Array.LOCALE); // returns ['a', 'á', 'z']
var newArrRev = arr.sort(Array.LOCALE | Array.DESENDING); // ['z', 'á', 'a']

var arr = ['Á', 'z', 'a'];
var a = arr.sort(Array.LOCALE | Array.CASEINSENSITIVE); // returns ['a', 'Á', 'z']

Array.LOCALE

Scaleform version: 3.0.65

:
String.localeCompare
```

11 문자열 클래스 확장

localeCompare() 메소드

```
public localeCompare(other:String [, caseInsensitive:Boolean]) : Number
```

Scaleform version: 3.0.65

Parameters

```
other:String -
```

`caseInsensitive:Boolean` -

$$\vdots$$

Array.sort

12 Video 클래스 확장

```
        . Scaleform  
NetStream
```

clipRect 속성

```
clipRect: flash.geom.Rectangle [read-write]
```

Scaleform version: 3.0.70

```
:  
video.clipRect = new flash.geom.Rectangle(10, 20, 100, 100);
```

13 전역 확장

noInvisibleAdvance 속성

noInvisibleAdvance : Boolean

Scaleform version: 2.1.52

```
True
    SWF
(
    Scaleform
    :
    MovieClip.noAdvance
```

imecommand 함수

imecommand(command:String, parameters:String) : Void

Scaleform version: 2.1.50

```
fscommand
    Scaleform IME
    . Scaleform
```

getIMECandidateListStyle () 함수

public function getIMECompositionStringStyle(categoryName:String): Object

Scaleform version: 2.1.50

```
IME
    setIMECandidateListStyle
```

반환값

Object –

```
:  
setIMECandidateListStyle  
TextField.disableIME
```

setIMECandidateListStyle () 함수

```
public function setIMECandidateListStyle(styleDescriptor:Object)
```

Scaleform version: 2.1.50

styleDescriptor

```
styleObject.textColor = 0x5EADFF;  
styleObject.selectedTextColor = 0xFFFFFFFF;  
styleObject.fontSize = 20;  
styleObject.backgroundColor = 0x001430;  
styleObject.selectedTextBackgroundColor = 0x2C5A99;  
styleObject.indexBackgroundColor = 0x12325A;  
styleObject.selectedIndexBackgroundColor = 0x2C5A99;  
styleObject.readingWindowTextColor = 0xFFFFFFFF;  
styleObject.readingWindowBackgroundColor = 0x001430;  
styleObject.readingWindowFontSize = 20;
```

textColor : Number

selectedTextColor : Number

fontSize : Number

backgroundColor : Number

seletedTextBackgroundColor : Number

indexBackgroundColor : Number

selectedIndexBackgroundColor : Number

readingWindowTextColor : Number

readingWindowBackgroundColor : Number

readingWindowFontSize : Number

0xRRGGBB . RR 16 [0..255],
BB 16 [0..255] GG 16 [0..255] .
:
getIMECandidateListStyle
TextField.disableIME

14 3Di 클래스 확장

Scaleform 3Di Scaleform AS2 . 3Di
ActionScript .

Scaleform version: 3.2.82

_z

Z () . 0 .

_zscale

Z . 0 .

_xrotation

X() . 0 .

_yrotation

Y() . 0 .

_matrix3d

16 float (4 x 4) 3D . NULL
3D 2D .

_perspfov

(Field Of View) , 0 180
FOV (55) .

gfxExtensions ActionScript true

15 표준 메소드 및 이벤트 핸들러 익스텐션

Key 클래스

```
Key.getCode(controllerIdx:Number)  
Key.getAscii(controllerIdx:Number)  
Key.isDown(controllerIdx:Number)  
Key.isToggled(controllerIdx:Number)
```

```
        /                               Key                               .  
controllerIdx          0          .
```

```
Key.onKeyDown(controllerIdx:Number)
```

```
Key          onKeyDown          /  
        .
```

Selection 클래스

```
Selection.setFocus(ch:Object, controllerIdx:Number)  
Selection.getFocus(controllerIdx:Number)  
Selection.getBeginIndex(controllerIdx:Number)  
Selection.getEndIndex(controllerIdx:Number)  
Selection.setSelection(start:Number, end:Number, controllerIdx:Number)  
Selection.getCaretIndex(controllerIdx:Number)
```

```
        /                               Selection                               .  
controllerIdx          0          .
```

```
Selection.onSetFocus(old:Object, new:Object, controllerIdx:Number)
```

Selection onSetFocus /

.

MovieClip/Button/TextField

```
MovieClip.onSetFocus(old:Object, controllerIdx:Number)
MovieClip.onKillFocus(new:Object, controllerIdx:Number)
Button.onSetFocus(old:Object, controllerIdx:Number)
Button.onKillFocus(new:Object, controllerIdx:Number)
TextField.onSetFocus(old:Object, controllerIdx:Number)
TextField.onKillFocus(new:Object, controllerIdx:Number)
TextField.onChangeed(controllerIdx:Number)
```

MovieClip/Button/TextField /

.

System.capabilities

System.capabilities.numControllers – controller .